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FORGE WORLD Due-Diligence Dossier

Executive verdict and strongest reformulation

FORGE WORLD is worth pursuing, but not in the form implied by the phrase "a computational model of what the physical world is capable of building." That formulation overpromises observability. No public-data system can know, at global component/process/factory resolution, the true available capacity, yield, qualification status, machine-tool dependencies, tacit know-how, supplier relationships, proprietary bills of process, inventory, uptime, or redesign options of the industrial economy. Trade databases, input-output tables, patent data, facility databases, company disclosures and industrial-capability maps each observe only fragments of that state. [1]

The strongest defensible reformulation is:

> **FORGE WORLD should be an evidence-backed industrial buildability compiler and transformation optimizer: a system that converts a physical objective into alternative technical architectures, closes their production dependencies, identifies missing capabilities and capacity, schedules the transformation required to create them, tests the plan against shared-resource and geographic constraints, and produces an auditable proof of what is feasible, infeasible, uncertain, or unknowable under explicit assumptions.**

That is importantly different from a supply-chain map. Existing products such as Altana, Interos, Everstream, Resilinc and Sourcemap are already sophisticated at mapping, inferring and monitoring multi-tier supply networks; Kinaxis and anyLogistix already perform sophisticated planning, optimization or simulation over operational supply networks. [2] FORGE WORLD becomes genuinely differentiated only if it solves the harder **forward transformation** problem:

objective → valid product architecture → production recipes → capability closure → current capacity → missing capacity/capability → projects and lead times → qualification/ramp → competing demand → imports/license/invention/substitution choices → shock-resilient plan → feasibility proof.

IndiaBUILD is particularly important because it independently validates a large part of the *capability graph* intuition. As of August 21, 2026 it reported 7,047 mapped industrial capabilities and 3,874 assessed capabilities, using status categories ranging from "No capability" and "Emerging" through "Producing (licensed)," "Producing (own tech)" and "Competitive." It explicitly links upstream and downstream capabilities, builders and policy, and identifies areas such as machine tools, semiconductors and precision manufacturing among Indian gaps/chokepoints. [3] FORGE should learn heavily from this ontology rather than reinventing the concept of industrial depth.

But IndiaBUILD is not, from its public presentation, a quantitative capacity-and-time optimizer. Its rare-earth representation is nevertheless an excellent model of the conceptual depth FORGE needs: it follows the chain

through deposits, separation, metals, master alloys and magnets, while also recognizing recycling and rare-earth-free pathways. [3] That is much closer to the correct abstraction than a simple HS-code supply chain.

The most important conclusion from this investigation is therefore:

The graph is not the product. The solver is not the product. The data lake is not the product. The product is the auditable connection between engineering reality, industrial capability, time, capacity, alternatives and evidence.

A second conclusion is equally important:

> **FORGE should never claim "this target is feasible" without specifying the product architecture, time horizon, capacity assumptions, sourcing rules, substitution set, qualification requirements, market-allocation assumptions and unresolved unknowns that make the conclusion true.**

The appropriate result classes are consequently not just *yes* and *no*, but:

Result	Meaning
FEASIBLE	At least one validated architecture satisfies the modeled engineering, capacity, timing and policy constraints.
CONDITIONALLY FEASIBLE	Feasibility depends materially on explicit assumptions, unverified capacity or projects that have not yet matured.
INFEASIBLE UNDER MODEL	No modeled architecture meets the objective; the solver can expose binding constraints or an irreducible conflicting set.
NOT ADJUDICABLE	Missing evidence is sufficiently important that a defensible conclusion cannot be reached.

That last category is essential. A system that says **UNKNOWN** at the right moment is strategically more useful than one that hallucinates a complete industrial base.

What is already available versus what is potentially novel

Capability	Existing state of the art	FORGE opportunity
Country/product relatedness	Harvard Atlas/product-space methods; WIPO's 2026 Innovation Capabilities Navigator covers 193 economies and over 2,500 capability fields across science, technology, entrepreneurship and production. [4]	Use as priors for what a country may be adjacent to, not proof that a component can be produced.

Capability	Existing state of the art	FORGE opportunity
National capability mapping	IndiaBUILD already implements a deep industrial-capability catalog with maturity distinctions and upstream/downstream relationships. [3]	Add quantitative recipes, capacity, investment, time, qualification and optimization.
Global trade structure	UN Comtrade, OECD ICIO/TiVA, Eora, WITS/ GTAP. [5]	Bridge aggregate trade/ economic categories to engineering objects with explicit uncertainty.
Multi-tier supply graph	Altana, Interos, Everstream, Resilinc, Sourcemap. [6]	Create *counterfactual* networks for industrial objectives that do not yet have a supply chain.
Operational planning/digital twin	Kinaxis and anyLogistix model scenarios, production/logistics effects and network optimization/simulation. [7]	Model the creation of new industrial capabilities, not primarily operation of an existing enterprise network.
Critical-mineral intelligence	IEA, USGS, EU JRC/RMIS and geological agencies provide increasingly sophisticated resource, processing and dependency analysis. [8]	Connect mineral constraints all the way to redesignable products and capital-equipment requirements.
Macro shock modeling	Production-network economics, MRIO and CGE systems already address propagation and equilibrium effects. [9]	Combine macro feedback with physical recipes, inventories and explicit bottleneck capacity.
Transformation planning	Capacity-expansion and dynamic input-output methods exist in individual sectors. [10]	Generalize the concept across manufacturing while preserving engineering detail.

The novelty thesis is therefore **credible but narrow**. FORGE WORLD would not be novel because it has a graph of products, companies, mines or ports. It would be novel if it can reliably answer questions of the form:

> "Given what exists now, these technical alternatives, these permitted imports, these project lead times, and these uncertainties, here are the minimum capability additions required; here is when they could become qualified; here are the constraints that bind; here are the substitutions that remove them; and here is the evidence behind every conclusion."

That is a much harder and more valuable project.

World model, ontology, and mathematical architecture

The central architectural mistake to avoid is attempting to choose one grand mathematical framework. Input-output economics cannot represent a precision grinding operation. A BOM cannot clear the copper market. A discrete-event factory simulation cannot model international commodity competition. A CGE model cannot determine whether India can produce a humanoid-grade crossed-roller bearing. An LLM cannot safely infer any of these from text.

FORGE therefore needs a **layered model with interfaces between layers**.

Recommended conceptual stack

Layer	Primary objects	Function	Appropriate methods
Evidence and identity	claims, sources, observations, versions, organizations, facilities, classifications	Establish what is actually known	provenance graph, entity resolution, temporal/ versioned database
Product architecture	product variants, functions, subassemblies, components, performance envelopes	Describe *what* can satisfy the objective	configuration models, BOMs, systems engineering, constraint satisfaction
Production recipe	process steps, material grades, yields, scrap, energy, tooling, equipment	Describe *how* each item can be made	bill-of-process, process planning, PPR ontology
Capability	process capability, quality envelope, IP, tooling, skills, qualification	Determine *who/ where can actually execute a recipe*	capability graphs, maturity vectors, evidence rules
Assets and capacity	mines, plants, lines, machines, projects, throughput, utilization, yields	Bound quantities over time	capacity-constrained network flow, LP/MILP, capacity expansion
Logistics and geography	sites, regions, ports, corridors, transit times, route capacities	Move physical inputs through space	min-cost flow, multimodal network optimization, geospatial analysis
Markets and competing demand	commodity pools, sector demand, inventories, prices, trade	Prevent target industries from claiming unlimited resources	allocation models, partial equilibrium, MRIO; CGE selectively

Layer	Primary objects	Function	Appropriate methods
Transformation and resilience	new projects, substitution, redesign, shocks, policy, investment portfolios	Determine how the world can change	dynamic MILP, scenario optimization, robust/ stochastic optimization

The manufacturing core should be based on an explicit **Product–Process–Resource** pattern. Manufacturing-ontology research repeatedly converges on products, processes, resources, measurements and constraints as fundamental concepts, while capability ontologies such as MaRCO are specifically intended to formalize what manufacturing resources can do and under what constraints. [11]

The minimal useful ontology should include the following distinct entity types rather than collapsing them into generic “nodes”:

Entity	Why it must be separate
Objective	Quantity, date, geography, performance requirements and policy constraints.
ProductVariant	There is no canonical “humanoid,” “solar panel” or “satellite.” Each architecture has different requirements.
FunctionalRequirement	Allows substitutions to be evaluated against required performance rather than matching part names.
Component / Subassembly	Physical decomposition.
MaterialGrade / ChemicalSpec	“Steel,” “aluminium” or “copper” is usually too coarse; many processes require specific alloys, purity or morphology.
ProductionRecipe	A validated alternative route from inputs to an output.
ProcessStep	Machining, heat treatment, epitaxy, sintering, coating, assembly, inspection, etc.
ResourceClass	What a process requires: equipment class, workforce capability, cleanroom class, metrology, utilities.
Machine / Tool / Fixture	Capital good that may itself be recursively produced.
Facility / ProductionLine	Location-bound physical production asset.
CapacitySegment	Quantity, unit, period, compatible products, utilization, yield and effective dates.
Project	Future mine, fab, plant, line or infrastructure addition with state and schedule.
Organization	Owner, operator, supplier, licensor, regulator or certifier.
CapabilityClaim	Evidence-backed assertion that an organization/geography can execute a process to a specified envelope.

Entity	Why it must be separate
Qualification	Customer, regulatory or sector-specific acceptance; distinct from technical manufacturability.
IP / License / TechnologyRight	Whether the process can legally or independently be reproduced.
Skill / KnowledgeAsset	Tacit or explicit production know-how where it materially gates ramp.
GeographicNode / Corridor	Place and transport topology.
DemandPool / Inventory	Competing consumption and buffers.
Policy / Restriction	Tariff, export control, subsidy, local-content rule, sanction or eligibility constraint.
Shock	Time-bounded change to one or more underlying constraints.
AlternativeArchitecture	Technically validated substitution/redesign path.
EvidenceRecord / Claim	Provenance-bearing basis for every state variable.
ModelRun	Immutable configuration, assumptions, solver version and outputs.

Critical relationships include:

`decomposes_into`, `satisfies_function`, `produced_via`, `requires_input`, `consumes_energy`, `requires_capability`, `requires_equipment`, `requires_qualification`, `has_capacity`, `located_at`, `owned_by`, `licensed_from`, `supplied_by`, `transported_via`, `competes_for`, `substitutable_with`, `gated_by`, `enabled_by_project`, `affected_by_shock`, and `evidenced_by`.

The graph must be **typed and versioned**. An untyped knowledge graph in which every edge merely means "depends on" is too ambiguous for computation.

Which intellectual frameworks belong where

Industrial ecology and material-flow accounting are valuable for physical conservation, environmental burdens and macro sanity checks. Eurostat's material-flow methodology, for example, records physical flows entering and leaving economies by material categories and physical trade, but its aggregate tonne-level accounts are not substitutes for product-level production recipes. [12] FORGE should use MFA/LCA as a **background physical-accounting layer**, not the main feasibility engine.

Ecoinvent is potentially valuable for background energy, water, materials, waste and environmental process intensities; its current database contains more than 26,000 process/activity datasets and is explicitly designed as background lifecycle inventory data. But integrating ecoinvent into software requires an appropriate developer license, making it a licensed enrichment layer rather than an assumed freely redistributable core dataset. [13]

Economic complexity is highly useful for **capability priors and industrial adjacency**. The classic product-space work showed empirically that products co-occur in national export baskets in structured ways, while the WIPO/Harvard capability system now extends relatedness thinking across production, science, technology and entrepreneurship. [14] It should not be interpreted as evidence that a country can make a specific qualified part at a required tolerance and annual volume.

Input-output and MRIO models are important for aggregate indirect demand and macroeconomic coupling. OECD's ICIO system covers inter-country production, consumption, investment and trade; its 2026 constant-price extension covers 80 economies plus the rest of the world, 50 industries and 1995–2024, with recent periods partly projected. Eora similarly supplies a high-resolution MRIO time series across roughly 190 countries. [15] These classifications are still orders of magnitude too coarse for humanoid actuators, wafer processes, gear grinding or specialty chemicals.

CGE is useful much later when a FORGE scenario becomes large enough to alter wages, exchange rates, sector output, trade patterns and demand. GTAP is a mature multi-region, multi-sector CGE system, explicitly assuming such structures as perfect competition and constant returns in its standard model. [16] Those assumptions make it an economic response layer, not a manufacturing-feasibility oracle.

Production-network science is directly relevant to shocks. Research in this literature demonstrates why idiosyncratic supplier/sector shocks can propagate through production linkages and why network structure can amplify disturbances. [17] FORGE should combine that insight with explicit material and capacity constraints rather than assuming proportionate downstream price effects.

Discrete-event simulation should be used **selectively**: plant queues, ports, inventory transients or ramp behavior for a small number of critical systems. Attempting to run a global factory-level DES would require data that do not exist and would produce impressive-looking false precision.

System dynamics can be useful for high-level feedback exploration, but is too aggregated for the central buildability question.

Agent-based modeling belongs later, for behaviors that cannot reasonably be represented through optimization—supplier entry/exit, strategic inventory, procurement behavior, sanctions evasion or endogenous firm responses. It is not a V0 requirement.

Causal models should estimate particular empirical parameters—effects of outages, policy, lead times or learning—rather than serving as the entire world representation.

Bayesian models and interval/probabilistic uncertainty should be first-class because FORGE will routinely operate on incomplete industrial evidence. They should quantify uncertainty around parameters and latent capabilities, while never converting absence of evidence into fictitious precision.

The core optimization formulation

A high-value FORGE query naturally becomes a **time-indexed, capacity-constrained, multi-recipe network design problem with discrete investment decisions**.

Conceptually, the decision variables include:

- production quantity of each recipe at each location and time;
- domestic versus imported sourcing;
- activation of new facilities/projects;
- equipment acquisition or domestic manufacture;
- technology-license acquisition;
- product architecture and substitution choices;
- inventories and transport flows;
- optional qualification or R&D projects.

A simplified effective-capacity representation should be:

$$C^{eff}_{f,p,t} = C^{nameplate}_{f,p,t} \times U_{f,t} \times Y_{f,p,t} \times A_{f,t} \times Q_{f,p,t}$$

where U is achievable utilization, Y is yield, A is uptime/availability and Q is the fraction of capacity actually qualified/eligible for the relevant product. This prevents a nominal 10,000-unit line from being treated as 10,000 deliverable units immediately after commissioning.

The solver should support multiple objectives rather than pretending everything reduces cleanly to dollars:

$$\min \left[\text{Capex}, \text{Time}, \text{Import dependence}, \text{Concentration risk}, \text{Expected disruption loss}, \text{Emissions}\right]$$

The preferred UI concept later is therefore a **Pareto frontier**, not a single arbitrary resilience score.

For the question “What is the smallest set of investments required to remove India’s most dangerous industrial dependencies?” fixed-charge investment variables make MILP particularly natural. FORGE can choose among a few large enabling investments and calculate downstream closure effects.

For impossible plans, the solver should expose the **binding constraints and minimal infeasible set**, turning failure into useful industrial intelligence:

> “The 2030 target fails under this architecture because qualified reducer capacity reaches only X, the earliest validated new line is after the deadline, and importing the missing quantity violates the no-China constraint.”

That is far more useful than a risk heatmap.

Machine-to-machine recursion

The proposed recursion is real and strategically important. Dynamic input-output models have long represented capital formation and production-capacity growth, including explicit links between investment goods and future capacity. [10] But a literal recursive decomposition of every capital good into every part of every machine is computationally and epistemically hopeless.

FORGE should implement **lazy strategic recursion**:

Product → process recipe
→ equipment/resource requirements
→ test whether the equipment dependency is strategically material
→ recurse into the equipment only when it matters.

A CNC grinder therefore begins as an externally purchasable capital resource. FORGE expands it into spindles, servomotors, bearings, controllers, encoders, metrology, abrasives, tool materials and their production capabilities only when one of those dependencies becomes binding.

The recursion trigger should depend on a **strategic-closure score**, combining factors such as supplier concentration, import-denial scenario, lead time, replacement difficulty, capital-equipment bottleneck centrality, domestic policy constraint and solution shadow price.

Stop recursion when:

1. the object can be reliably purchased under the scenario;
2. its contribution to feasibility/risk is below a defined threshold;
3. the model reaches a well-supplied commodity or standardized resource class;
4. uncertainty exceeds the value of further decomposition; or
5. a user policy boundary explicitly allows import dependence at that level.

This preserves the important insight—**machines that build machines matter**—without pretending FORGE must first reconstruct civilization atom by atom.

The IEA's recent critical-minerals work provides a striking reason to include equipment depth: it notes that diversification can itself be constrained by specialized processing equipment, workforce and know-how, with only a handful of suppliers outside the dominant producer for some graphite and rare-earth processing equipment. [18] In other words, adding “build a refinery” is not enough if the machines and know-how required to operate that refinery constitute the actual bottleneck.

Capability, industrial time, shocks, substitution and geography

Industrial capability must be a vector, not a flag

IndiaBUILD's explicit distinction among emerging capability, licensed production, own-technology production and competitive production is directionally correct and should be adopted rather than reduced to `country_has_capability = true`. [3] NASA TRLs similarly demonstrate why technology maturity is not binary, while U.S. defense acquisition practice explicitly distinguishes production readiness from technological maturity and evaluates such issues as high-risk manufacturing processes, yield, tools, test equipment and long-lead items before production. [19]

FORGE should store a **capability vector**.

Dimension	Example questions
Technical demonstration	Has the process/product ever been demonstrated at required performance?
Manufacturing maturity	Prototype, pilot, low-rate, stable series production or high-volume production?
Capacity	Qualified effective annual output today and planned?
Yield/process control	Is production repeatable at acceptable scrap/yield?
Scale-up maturity	Has throughput comparable to the objective ever been demonstrated?
Design ownership	Is the design domestic, licensed, jointly developed or wholly external?
Process ownership	Can the production process be reproduced and modified independently?
Tooling autonomy	Are critical tools, dies, fixtures and process recipes domestically controlled?
Capital-equipment dependency	Can production continue if imported machinery/spares/service disappear?
Precursor dependency	Are critical alloys, chemicals, substrates, magnets or feedstocks imported?
Qualification	Is the supplier qualified for the particular customer/industry/application?
Metrology/test	Can required quality be measured and certified domestically?
Workforce/know-how	Does tacit process knowledge exist at adequate scale?
Lead time	How rapidly can additional qualified output be created?
Cost position	Is production merely possible, commercially viable, or globally competitive?
Concentration/resilience	How many independent sites/suppliers/regions support the capability?

A convenient user-facing label can still be derived—**Imported only**, **Assembly**, **Licensed production**, **Independent production**, **Scaled**, **Competitive**, **Resilient**—but the solver should operate on the vector, not the label.

This also fixes the ambiguous concept of “domestic.” Consider India's semiconductor program. As of July 15, 2026, the government reported 12 approved manufacturing projects: one silicon fab, one compound-semiconductor fab with packaging, one micro-LED/GaN fab with packaging and nine ATMP/OSAT facilities. [20] The Tata silicon fab was approved with a stated capacity around 50,000 wafer starts per month and a technology partnership with Taiwan's PSMC. [21] FORGE must therefore separately encode **geographic**

domesticity, **technology ownership**, **commissioning state** and **production maturity**. “A semiconductor fab exists in India” cannot safely substitute for all four.

Time is a state machine, not a capex field

Industrial capacity should pass through states such as:

concept / announced → engineering & permitting → financed/FID → procurement → construction → mechanical completion → commissioning → process qualification → customer/certification qualification → ramping → mature → impaired/retired.

This distinction is not cosmetic. Mineral supply illustrates the extreme case: the IEA has found that major mines coming online in a historical sample required more than 16 years on average from discovery to first production, with exploration/feasibility itself taking over a decade and construction adding several more years. [22] Semiconductor production has a different time structure: SIA describes wafer fabrication cycles of roughly 12 weeks on average and longer for advanced processes, followed by additional assembly/test and yield-ramp considerations. [23] Neither can be represented as “allocate money in 2028; receive capacity in 2028.”

A project object therefore needs:

`state`, `state_date`, `earliest_start`, `lead_time_distribution`, `commissioning_distribution`, `design_capacity`, `ramp_curve`, `expected_yield_curve`, `qualification_gate`, `required_capital_goods`, `required_infrastructure`, `permitting_risk`, `dependency_projects`, and `evidence`.

Ramp should normally be modeled piecewise rather than assuming full nameplate on day one:

$$C_t^{available} = C^{nameplate} \cdot r(t - t_0) \cdot U_t \cdot Y_t \cdot Q_t$$

where $r(\cdot)$ is a ramp curve from zero toward mature throughput.

Figure's own 2026 manufacturing disclosures illustrate why these terms matter even for a modern robotics company: it reported increasing Figure 03 output from one robot per day to one per hour in under 120 days, while reporting end-of-line first-pass yield above 80%, a battery-line first-pass yield of 99.3%, and more than 9,000 actuators produced across more than ten SKUs. Those are company-reported figures, not independent measurements, but they demonstrate that **throughput, yield, SKU mix and ramp are separate state variables**. [24]

Learning curves should alter labor hours, yield, cycle time and cost where empirical parameters exist. They should not automatically increase fundamental physical capacity without the required machinery.

Qualification needs its own DAG. DoD production-readiness guidance explicitly treats production tooling, process maturity, high-risk/low-yield manufacturing and long-lead items as distinct production-readiness concerns. [25] A supplier that can machine a gear is not necessarily a supplier qualified to produce millions of low-noise robot reducer gears.

Shock architecture

A shock should **modify constraints**, not directly modify every downstream output.

FORGE should classify shocks into:

Shock class	Direct model action
Physical supply outage	Reduce facility/resource capacity or yield.
Mine/plant closure	Set affected capacity unavailable over event interval.
Logistics closure	Disable route or reduce throughput; increase time/cost on alternatives.
Export control/sanction	Change supplier/product eligibility by origin and destination.
Tariff	Add border cost; potentially trigger alternate sourcing.
Commodity-cost shock	Change exogenous offer curve or marginal input cost, then allow downstream optimization.
Demand shock	Change competing demand pool.
Electricity shortage	Reduce available power or apply curtailment schedules/prices by location.
Currency shock	Change landed-cost and financing relationships.
Interest-rate shock	Change project financing/carrying costs and potentially investment selection.
Factory fire/natural disaster	Site-specific capacity impairment plus restoration state.
New mine/fab/refinery	Add a project and its realistic state/ramp rather than instant supply.
New technology	Add an alternative recipe with its maturity, cost and qualification status.

The sequence should be:

shock → direct changes to capacity/routes/eligibility/demand/cost
→ inventory drawdown and reallocation
→ constrained production re-optimization
→ substitution/redesign if permitted
→ changed utilization and scarcity values
→ price/market feedback
→ indirect macro effects where relevant.

This is more defensible than “copper doubled, therefore every downstream product's copper-related cost doubles.” Production-network literature shows that propagation depends on network topology and input relationships rather than uniform multipliers. [26]

For short shocks, **inventory and order timing** become essential. This layer is missing from the original concept and should be added. Two companies with identical annual consumption can have radically different production outcomes if one carries four months of a constrained component and the other operates just-in-time.

For large, sustained commodity shocks, prices should increasingly become endogenous. V0 can use supply curves, elasticity envelopes and explicit allocation constraints for a handful of strategic commodities. Later versions can couple the engineering model to partial-equilibrium models. MRIO provides indirect-demand effects; CGE should be reserved for cases large enough to shift the broader economic equilibrium. [27]

Technology substitution and redesign

Substitution is where FORGE can become far more powerful than a static supply-chain model—but also where an LLM can destroy credibility.

The LLM must never be allowed to insert a substitution directly into the solver.

A substitution should be a versioned **validated alternative recipe** with:

- functional equivalence envelope;
- changed BOM;
- changed BOP;
- performance penalties/advantages;
- manufacturing capability requirements;
- new tooling;
- technology/manufacturing maturity;
- qualification and certification changes;
- capex and transition costs;
- lead time;
- supplier availability;
- evidence.

Examples should look like:

`NdFeB PMSM architecture` ↔ `alternative rare-earth-reduced motor architecture`

not:

`rare earth unavailable → AI says use induction motor → feasible`.

The model must establish that the replacement still meets torque density, speed, thermal envelope, efficiency, size, controllability, noise, duty cycle and lifetime requirements.

Mathematically, architecture/recipe choice belongs naturally in MILP:

$$\sum_{r \in R_p} z_{p,r} = 1$$

for mutually exclusive recipes, with recipe-specific material, process, performance and transition constraints.

Substitution can also occur at different semantic levels:

material substitution → component substitution → process substitution → subassembly redesign → system-architecture redesign.

These must not be conflated. A constrained magnet might be solved through a different magnet grade, a different motor geometry, a different motor technology, a gearbox-ratio change, or a wholesale actuator redesign. Each has different consequences.

Recycling should enter as a time-dependent secondary-production recipe, including collection efficiency, recovery yield and end-of-life stock availability. EU JRC work on critical-material recovery demonstrates why theoretical material content and actually recoverable material are not equivalent; collection and treatment losses can be substantial. [28]

Competing demand

The original concept correctly recognizes that the target industry cannot simply seize world supply. However, implementing this with a full global general-equilibrium model in V0 would be a mistake.

A staged approach is preferable.

V0: use observed/global production, baseline demand and conservative reserve constraints for strategic commodities. Define:

$$D_{m,t}^{FORGE} + D_{m,t}^{baseline} \leq S_{m,t}^{available} + I_{m,t} + S_{m,t}^{new}$$

with alternative scenarios for how much baseline demand is realistically contestable.

Next: add explicit supply curves, substitution elasticities and allocation among several competing strategic industries.

Later: link high-impact scenarios to MRIO or GTAP/CGE for broader income, trade and demand feedback. OECD ICIO and GTAP were built precisely for cross-sector and cross-border economic interdependence, but at much coarser sector resolution than FORGE's engineering graph. [29]

Important shared bottlenecks are not only commodities. FORGE should support competing demand for:

machine-tool hours, semiconductor wafer capacity by technology class, specialty chemical capacity, freight corridors, grid connections, qualified engineering labor, test facilities and specialized capital equipment.

That is one of the places where FORGE could substantially outperform commodity-only industrial models.

Geography

Global city-level precision is not defensible for V0.

The realistic geographic hierarchy is:

global → country → first-level administrative region where data support it → strategic facility/site nodes.

For India, state-level detail is defensible for selected industrial/electricity/manufacturing variables, supplemented by geocoded mines, ports, power assets, semiconductor projects and strategic industrial sites. MoSPI's ASI is the major official statistical source for registered manufacturing and currently covers roughly 2.6 lakh factories in the 2023–24 round; MoSPI has also released tools for accessing its microdata holdings. [30] It still cannot tell FORGE the tolerance, yield or qualification status of a particular reducer line.

Useful spatial layers already exist. Global Energy Monitor's integrated power datasets cover a very large global facility universe and are released under CC BY 4.0; OpenStreetMap provides road/rail/port and industrial geometries under the ODbL; UNCTAD publishes liner-shipping connectivity indicators; India's CEA provides monthly installed-capacity, generation and plant/project monitoring data. [31]

The geography engine should therefore model **networks and catchments**, not just points on a map:

facility → road/rail node → port → maritime route → destination port → industrial cluster.

Location constraints should include power, water where reliably measurable, transport, permitting, industrial land, infrastructure connection and hazards only when there is evidence. A beautiful city-level heatmap built from country-level proxies would be worse than no map.

Data, APIs and existing-system landscape

There is no single dataset capable of supporting FORGE. The project must be built around **data fusion with explicit resolution boundaries**.

The most damaging data-integration problem will be classification mismatch:

HS trade codes ≠ industrial products ≠ engineering parts ≠ manufacturing processes ≠ company product catalogs ≠ ISIC/NAICS sectors ≠ patent classes.

The Harvard Atlas itself performs substantial harmonization and reconciliation across trade classifications and partner reports, while OECD must reconcile heterogeneous national input-output systems to build ICIO. [32] FORGE will need its own many-to-many **classification crosswalk graph with confidence and provenance**. A one-to-one HS→component lookup is not technically credible.

Priority data/source matrix

Source	What it contributes	Access / legal posture	Main limitation for FORGE
IndiaBUILD	India capability ontology, upstream/downstream relationships, capability maturity, builders and policies; thousands of mapped capabilities. [3]	Public web access; bulk/API and commercial redistribution rights should be explicitly negotiated before product dependence.	Excellent capability structure; public presentation does not provide the quantitative factory-capacity/time model FORGE requires.
WIPO Innovation Capabilities Navigator	Capability profiles for 193 economies across >2,500 science, technology, entrepreneurship and production capabilities. [33]	Public WIPO service; embedding/bulk rights should be checked by dataset.	Latent/national capability signal, not production-line proof.
Harvard Atlas	Product-space/economic-complexity and trade-related capability signals. [34]	Downloadable research data; terms must be checked for product redistribution.	Export-based capability inference; coarse relative to engineering processes.
UN Comtrade	Bilateral merchandise trade by commodity/country/time; APIs and detailed downloads. Free API registration supports sizable but rate-limited queries. [35]	Public API/bulk access under UN Comtrade license.	Trade does not reveal producer, facility capacity, domestic value-added or technical process.
DGCI&S TradeStat	Official Indian merchandise-trade detail; current portal includes data through 2025–26 and updated 2026 monthly series. [36]	Public query access; redistribution conditions require legal review.	Customs classification ambiguity, re-exports/ reimports and unspecified partner categories can complicate inference. [37]
OECD ICIO / TiVA	Multi-country sector production, intermediate use, final demand, investment and trade; value-added decomposition. [38]	OECD data access; verify redistribution terms for a commercial integrated product.	Sector aggregation and publication lag make it a background model, not real-time capacity data.

Source	What it contributes	Access / legal posture	Main limitation for FORGE
Eora MRIO	High-resolution global MRIO with environmental/social satellite accounts for roughly 190 countries. [39]	Downloadable under Eora's access arrangements; commercial use/ redistribution must be reviewed.	Statistical/model reconciliation; far too coarse for components and processes.
UNIDO INDSTAT	Industrial statistics across manufacturing activities and countries, including long historical series. [40]	UNIDO now provides relevant database access free under Creative Commons terms.	Establishment/process detail remains limited relative to FORGE needs.
MoSPI ASI	Indian registered factory-sector structure, employment, output, inputs and related manufacturing variables; microdata available through MoSPI systems. [41]	Official data/reports are free; microdata use and redistribution rules must be respected. [42]	Does not expose proprietary BOM/BOP, exact products, tolerance or qualified capacity.
USGS MCS / Mineral Industry Surveys	World mine production, reserves/resources context, U.S. statistics, industry structure, monthly/quarterly mineral surveys for >90 nonfuel mineral commodities. [43]	Public USGS data products/CSV; verify individual dataset notices.	Refining/intermediate-product chains can be much less transparent than mine production.
IEA critical-minerals data	Mining/refining concentration, project pipelines, demand scenarios, policies and selected supply-chain risks. Current critical-minerals dataset was updated in July 2026. [44]	The cited dataset/ policy tracker are CC BY 4.0. [45]	Excellent strategic analysis but not component-level BOM data.
Global Energy Monitor	Geocoded power, oil/gas and related energy infrastructure plus ownership. [46]	CC BY 4.0. [47]	Energy-focused; not a universal manufacturing-facility registry.

Source	What it contributes	Access / legal posture	Main limitation for FORGE
Indian Bureau of Mines	Indian mineral production, prices/resources and Indian Minerals Yearbook; monthly mineral-production series are available through 2026. [48]	Public government publications; machine-readable availability varies.	Data granularity varies by mineral; downstream processing capacity is unevenly observable.
GSI	Geological/exploration information and mineral-target data. [49]	Government data portals; dataset-specific terms/access vary.	Geological resource ≠ economically viable mine ≠ refined material capacity.
CEA / GRID-INDIA	Installed generation, plant-level performance, transmission, demand, project monitoring and operational power-system data. [50]	Public Indian government/ operator data; Excel/ report formats frequently available.	Industrial-site power quality, connection queues and contractual availability require additional data.
UNCTAD shipping connectivity	Country/port liner-shipping connectivity and temporal changes. [51]	Public UNCTAD statistical access.	Connectivity indices are not vessel-level capacity or actual shipment routing.
OpenStreetMap	Roads, rail, ports and other infrastructure geography. [52]	ODbL; attribution and share-alike implications must be designed into the data architecture.	Completeness and tagging consistency vary geographically.
WIPO PATENTSCOPE	Patent documents and bibliographic/IP evidence; WIPO provides programmatic services for publication documents. [53]	API/data use subject to authorized-use terms.	Patent ownership or filing does not prove operational production capability.
EPO OPS / Espacenet	Global patent bibliographic, full-text, legal-status and family information; Espacenet covers over 150 million patent documents and is updated daily. [54]	OPS is free under fair-use limits up to 4 GB/week, with paid subscription options and specific terms. [55]	Same patent→capability inference problem; entity resolution is difficult.
ecoinvent	Process-level background materials, energy, water, emissions and waste inventories. [56]	Paid/licensed; software integration requires a developer license. [57]	Environmental/ background inventory, not facility capacity or industrial qualification.

Source	What it contributes	Access / legal posture	Main limitation for FORGE
World Bank WITS	Trade, tariff and non-tariff measure access combining sources including UN Comtrade, UNCTAD TRAINS and WTO data. [58]	Mixed underlying source terms; TRAINS access is free while some underlying tariff-line data have restrictions. [59]	Useful policy/trade layer, not engineering supply-chain detail.
Company filings and technical disclosures	Actual plants, investment, supplier/technology partnerships, product specs, sometimes capacity and ramp.	Public for filings but individual copyright/ data terms apply; commercial redistribution must be legally designed.	Self-reported, selectively disclosed, inconsistent and frequently future-looking.
Commercial shipment/ supply-chain datasets	Firm/facility/part relationships, shipments and inferred multi-tier networks. Altana, for example, combines customs, public, commercial and customer data while explicitly distinguishing observed and inferred links. [60]	Commercial contract; expect strict restrictions on redistribution.	Expensive and often opaque; still does not supply proprietary process recipes.

The missing “dataset” is actually the most important one: **validated foreground industrial recipes**.

No credible global public source provides:

> “Humanoid joint variant X requires motor architecture A, magnet grade B, bearing class C, reducer process D, these machine classes, these cycle times, these yields, this metrology and these qualification tests.”

FORGE must therefore create **domain knowledge packs** assembled from engineering literature, standards, manufacturer datasheets, patents, teardowns, supplier catalogs, company disclosures and expert review. Public macro datasets then enrich those packs rather than substitute for them.

Existing systems: what to copy and what remains

Altana may be the most important commercial reference for FORGE's epistemic architecture. Its public documentation describes a product network linking companies, facilities, products and shipments; it explicitly distinguishes observed from inferred information, attaches explanation/confidence to inference, and recommends human review. [61] FORGE should copy that discipline. What remains different is the proposed generation of a *future industrial system* from a target, including hypothetical new factories and missing capabilities.

Interos focuses on automatically mapping and monitoring supply ecosystems and producing continuous risk intelligence. [62] FORGE should learn from continuous entity/risk monitoring, but a supplier-risk platform is not the same as an engineering buildability solver.

Everstream presents network mapping/digital-twin capabilities joining companies, locations, shipments, lanes and materials with risk monitoring. [63] This invalidates any claim that "cross-layer supply-chain visualization" itself is novel.

Resilinc emphasizes multi-tier and part/site-level supplier mapping, combining supplier-validated data and AI. [64] The lesson is that supplier relationship truth often requires first-party validation; FORGE cannot infer everything from customs data.

Sourcemap is particularly relevant to provenance and chain of custody. It can attach transactional and documentary evidence such as purchase orders, invoices, bills of lading and certificates to supply-chain relationships. [65] A mature FORGE enterprise version should eventually support comparable evidence-backed private data.

Kinaxis demonstrates the value of concurrent scenario planning across suppliers, inventory, production and logistics rather than sequential siloed planning. [66] FORGE's transformation model should similarly recalculate the whole plan when a single constraint changes.

anyLogistix already combines network optimization and dynamic simulation for production capacity, inventory and logistics. [67] FORGE should not waste years rebuilding a general logistics simulator unless the capability is essential and not integrable.

WIPO/Harvard economic-complexity systems do the national-capability adjacency problem better than a FORGE team should initially attempt from scratch. [68] FORGE should ingest/derive compatible signals and concentrate its originality on linking adjacency to process requirements and transformation time.

IndiaBUILD should be treated as an intellectual peer/possible data partner rather than merely another source. Its capability granularity and maturity distinctions are already closer to FORGE's core problem than conventional economic datasets. [3]

EU JRC/RMIS and IEA show what high-quality strategic-material analysis looks like, including processing, diversification, recycling and substitution rather than mine reserves alone. [69]

GTAP/CGE and OECD ICIO already provide a mature answer to broad economic equilibrium and inter-industry linkage questions. [70] FORGE should couple to these approaches at the boundary rather than reproducing a macroeconomic model internally.

Defense-industrial analysis reinforces the centrality of capacity rather than nominal suppliers. RAND's work on contested logistics, for example, highlights industrial-base capacity, shared lines and surge-production limitations as important constraints. [71] U.S. DOE supply-chain work similarly treats manufacturing deployment and critical-juncture capacity as strategic investment problems. [72]

The competitive conclusion is uncomfortable but useful:

Almost every individual feature in the FORGE concept already exists somewhere. The opportunity lies in the cross-scale synthesis and in solving the inverse problem—what industrial transformation is necessary to make a new physical objective possible?

India and the one-million-humanoid benchmark

India is a strong flagship geography because official data are relatively rich at the macro/manufacturing/policy level while crucial component-level gaps remain—exactly the environment in which FORGE's epistemic discipline matters.

IndiaBUILD provides a deep starting capability graph and explicitly identifies machine tools, semiconductors and precision manufacturing among weak areas/chokepoints in its current national capability analysis. [3] MoSPI ASI provides registered-manufacturing structure; DGCI&S provides current trade intelligence; IBM and GSI cover minerals; CEA provides unusually detailed power-sector information; MeitY and the Ministry of Heavy Industries publish detailed project/program information. [73]

This combination is enough to build a valuable **India industrial observatory**. It is not enough to infer every relevant precision-production capability automatically.

Why the humanoid benchmark is an excellent stress test

There is no single canonical humanoid BOM.

Unitree's G1, for example, is about 35 kg and comes in configurations with 23–43 joint motors; its published specifications describe low-inertia internal-rotor PMSM joint motors, dual encoders and industrial-grade crossed-roller bearings. [74] Unitree's larger H1 is about 47 kg with an 864 Wh replaceable battery. [75] Figure 03 is reported at 61 kg with a 20 kg payload and five-hour runtime; Figure separately states that its F.03 battery is 2.3 kWh. [76]

Those differences prove why FORGE must begin from **reference architectures**, not from “a humanoid.”

For one million units/year:

- A G1-like 23–43-joint-motor architecture implies **23–43 million joint motors per year**. This is a deterministic derivation from Unitree's published configuration, not a forecast of what an Indian humanoid would use. [74]
- A battery range spanning the cited H1 and Figure examples implies roughly **0.864–2.3 GWh/year of finished robot battery energy capacity**. Again, this is an architectural bounding example, not a universal humanoid requirement. [77]
- A 35–61 kg robot envelope corresponds to approximately **35,000–61,000 tonnes/year of finished robot mass**, but this says essentially nothing about individual alloy, polymer, magnet or semiconductor requirements because public composition data are insufficient. [78]

This immediately reveals something important: for humanoids, the most interesting constraints may be **precision-component throughput, quality, tool capacity and supplier qualification**, rather than raw tonnage alone.

A useful FORGE reference architecture should decompose at least:

System	First dependency layer	Strategic recursion targets
Actuation	torque motor, reducer, bearings, encoder, housing, drive electronics	magnets/motor materials; precision gears/ flexsplines; grinding/forming; machine tools; metrology
Dexterous hands	miniature motors/transmissions, tactile sensing, electronics	micro-gears, force sensing, assembly yield, test
Perception	cameras, depth sensors/LiDAR where used, IMU	image sensors, optics, MEMS, packaging
Compute/ control	processors, memory, MCUs, power-management ICs, boards	semiconductor manufacturing/packaging, PCB/EMS, substrates
Power electronics	inverter/driver stages, MOSFETs/ IGBTs/GaN/SiC where architecture requires	wafers, packaging, thermal management
Energy storage	cells, BMS, busbars, pack structure, thermal/safety	active materials, cell production, pack qualification
Structure	aluminium/steel/composite/plastic parts	castings/extrusions, CNC, stamping, heat treatment, coating
Wiring	cable, connectors, harnesses	copper products, connector tooling, assembly/test
Thermal management	fans, heat spreaders, thermal interface materials	specialty materials and component production
Assembly/test	fixtures, calibration, end-of-line testers	automation equipment, metrology and diagnostics

Unitree's Dex3-1 hand is a useful reminder of how quickly the part count becomes more demanding than a simple robot BOM: its published hand uses seven active miniature joints plus tactile sensor arrays. [79] At one million robots, the exact choice of hand architecture could therefore dominate certain micro-actuator and sensor requirements.

Where India appears comparatively strong or is actively building capability

Electronics manufacturing and packaging: India has significant electronics assembly/manufacturing activity and is rapidly expanding semiconductor capacity through government-supported projects. As of July 2026, 12 semiconductor manufacturing projects had been approved under the first program, spanning a silicon fab, compound semiconductor, micro-LED/GaN and multiple ATMP/OSAT projects. [20] This is meaningful progress but should not be encoded as mature operating capacity until individual projects actually commission and ramp.

Semiconductor policy and design: Semicon 2.0 was approved on July 15, 2026 with a ₹1,27,500 crore outlay and explicitly includes chip design/IP as a pillar, after the government reported 105 startups already working on chip development. [80] This matters for the distinction between local assembly and technology ownership.

Battery industrialization: India's ACC PLI has a 50 GWh domestic-manufacturing target, of which 40 GWh had been awarded to four beneficiaries by early 2026. [81] Compared with the **derived** 0.864–2.3 GWh/year robot-battery envelope above, bulk cell-energy volume alone does not look like the obvious binding constraint. But awarded capacity is not commissioned capacity, and chemistry, power capability, form factor, safety, BMS and qualification may matter more than aggregate GWh.

Capital goods and machine-tool development: The Ministry of Heavy Industries operates the Phase II capital-goods competitiveness program, with a ₹1,207 crore financial outlay and projects spanning centers of excellence, common facilities, testing/certification and technology development; a dedicated Centre of Excellence on Machine Tools Design at IIT BHU/HMT is one example. [82] This is precisely the kind of capability-building evidence FORGE should model as *enabling projects*, not simply as industrial output.

Automotive and precision-volume manufacturing: India's large automotive/auto-components base is strategically relevant to motors, electronics, machining, castings, wiring and production engineering, and the current PLI-Auto scheme reported ₹44,326 crore of cumulative investment through March 2026. [83] But FORGE should not infer humanoid-grade components from automotive capability without product/process evidence.

Where serious dependency or uncertainty remains

Rare-earth permanent magnets are a demonstrated vulnerability. In July 2026 the Indian government stated that current domestic demand for sintered NdFeB rare-earth permanent magnets was being met through imports, while the new REPM scheme is designed to create 6,000 tonnes/year of integrated domestic capacity covering oxide→metal→alloy→finished magnet. [84] The Ministry reported receiving 20 bids for the program by August 13, 2026. [85]

This is almost a textbook FORGE case. The correct model status is not:

`India NdFeB capacity = 6000 t/y`.

It is:

`OBSERVED: scheme targets 6000 t/y`

+ `OBSERVED: tender/bid state`

+ `PROJECT: future candidate capacity`

+ `CURRENT CAPABILITY: import dependent`

+ `UNKNOWN: eventual successful project allocation, commissioning, product-grade compatibility, yield and qualification`.

The IEA's broader 2026 assessment also warns that rare-earth diversification outside the dominant producer remains constrained at refining, metal/alloy and finished-magnet stages, not merely mining. [86]

Precision reducers: public evidence is inadequate to confidently quantify India's current humanoid-grade harmonic/cycloidal reducer capacity. FORGE should mark this **UNKNOWN / requires supplier-level validation**, not infer it from generic gear production.

The evidence needed is highly specific: reducer architecture, torque class, backlash, efficiency, lifetime, noise, bearing integration, annual qualified capacity, proprietary tooth/flexspline processes, machine fleet, grinding/forming cycle time and metrology.

Encoders and force/torque sensors: India has electronics and instrumentation capability, but public macro datasets do not establish million-unit capacity for humanoid-grade absolute encoders or joint torque/force sensing. This should also remain an evidence-gathering target rather than a confident capability claim.

Crossed-roller and high-performance bearings: generic bearing production is not proof of robot-joint bearing capability. Unitree explicitly specifies industrial-grade crossed-roller bearings for its G1 joints, illustrating the application specificity. [74] FORGE would require supplier/product evidence and tolerances before declaring domestic closure.

Machine tools and metrology: these deserve deeper recursion because both IndiaBUILD and government capital-goods initiatives point toward precision machine-tool depth as strategically relevant. [87] The question is not whether India has CNC machines; it is whether enough appropriate high-precision grinding, turning, gear-making, calibration and measurement capacity exists, whether those machines depend on imported controllers/spindles/encoders/bearings and how quickly additional qualified capacity can be added.

Semiconductor independence: approved domestic fabs/OSAT significantly alter future geography of production, but FORGE should keep node/process/technology categories distinct. A robot may require advanced compute imported from one ecosystem, commodity MCUs from another, image sensors from another and power devices compatible with India's emerging compound-semiconductor capacity. A generic “semiconductor capacity” scalar would be nearly useless.

A defensible benchmark methodology

FORGE should not initially answer:

> “Can India make one million humanoids?”

It should answer a set of parameterized questions:

Reference architecture A: 35 kg-class, approximately 23 actuator joints, limited manipulation.

Reference architecture B: 50–60 kg industrial humanoid, approximately 30–40 powered joints, dexterous hands.

Reference architecture C: alternative actuator/motor design reducing rare-earth dependence.

For each, produce:

1. annual part/material/process requirements;
2. current evidence-backed Indian capability;
3. effective domestic capacity range where known;
4. currently importable capacity and concentration;
5. 2030 project pipeline;
6. missing capability projects;
7. machine/tool dependencies;
8. qualification timeline;
9. minimum new investment portfolio;
10. earliest feasible production ramp;
11. resilience under selected shocks;
12. unresolved unknowns and their decision importance.

The benchmark's likely **research bottlenecks**, based on present evidence, are therefore more interesting than a single 2030 yes/no answer:

Area	Present epistemic status
Bulk structural metals	Data comparatively strong at material/trade level; product-grade allocations still need recipes.
Battery energy volume	Macro capacity/project evidence fairly strong; exact compatible cell/pack capability requires architecture-specific validation.
Semiconductor projects/ packaging	Project data strong; future operating/yield and specific chip compatibility uncertain.
NdFeB magnets	Current dependence and future policy pipeline unusually well evidenced. [88]
Motors	Architecture can be characterized; Indian volume/capability needs supplier-specific evidence.
Precision reducers	Major evidence gap.
Encoders / torque sensors	Major evidence gap at required performance and volume.
Robot-specific bearings	Major evidence gap.
Precision machine-tool hours	Major evidence gap despite broader evidence that capability depth matters.

Area	Present epistemic status
Metrology / test / calibration	Sparse publicly standardized capacity information.
Assembly/EMS	Broad capability exists, but target-specific cycle time/yield must be modeled.
Qualification/ramp	Almost entirely company/project-specific.

That is not a failure of the benchmark. It identifies exactly what FORGE must be able to represent.

Provenance, uncertainty, technical risks and missing dimensions

The proposed epistemic classes are directionally correct, but **OBSERVED** is too broad. A government announcement of a proposed 6,000 t/year plant and a meter reading showing 6,000 tonnes actually produced are both “observed documents” but have completely different epistemic implications.

Use the following hierarchy:

Class	Meaning
MEASURED	Direct measurement/statistical observation under a documented methodology.
REPORTED	Assertion by company, government, researcher or other source.
DERIVED	Deterministic calculation from identified upstream claims.
ESTIMATED	Statistical inference or imputation from evidence.
MODELLED	Solver/simulation result conditional on model and inputs.
ASSUMPTION	Explicit analyst/model assumption.
SCENARIO	User-defined counterfactual.
UNKNOWN	No adequate evidence.
CONFLICTED	Material credible sources disagree.

The **epistemic class and confidence score must remain separate**. A company-reported capacity can be high-confidence evidence that the company made the statement while still being weaker evidence that the capacity will actually be realized.

Altana's explicit observed-versus-inferred distinction is an excellent commercial precedent. [60] W3C PROV provides a mature domain-agnostic model for entities, activities, agents and derivation relationships and is specifically designed to be specialized for domain-specific provenance. [89] FORGE should borrow that conceptual structure rather than inventing an idiosyncratic lineage system.

Every claim should carry, at minimum:

`claim_id`
`subject`
`predicate`
`value + unit/distribution`
`epistemic_class`
`source_id`
`source_type`
`publication_date`
`effective/event_date`
`retrieval_date`
`valid_from / valid_to`
`geographic_scope`
`technical_scope`
`extraction_method`
`confidence_dimensions`
`license`
`derived_from[]`
`review_status`

Every model result should similarly expose its dependency DAG.

A user clicking:

> "India requires 26 million imported actuator components in 2029"

should be able to trace:

model output → chosen robot architecture
→ unit actuator count
→ target volume
→ domestic-capacity assumption
→ project ramp assumptions
→ source documents
→ unresolved uncertainty.

Confidence should be multidimensional

A single 0.82 confidence number invites false interpretation. Track at least:

source authority, directness, corroboration, recency, geographic specificity, technical specificity, sample/completeness, and model sensitivity.

The most powerful additional feature would be **value of information**:

> Which unknown fact, if resolved, is most likely to change the investment decision?

If a reducer-capacity uncertainty changes the result from feasible to infeasible but a connector-cost uncertainty has almost no effect, FORGE should explicitly prioritize verification of the reducer evidence.

This transforms uncertainty from a disclaimer into an industrial-research agenda.

Major conceptual corrections

Trade is evidence of flows, not capability. A country can export a product after importing most high-value components, re-export goods, perform final assembly, or host foreign-owned licensed production. UN Comtrade and TradeStat are indispensable but cannot answer technical independence alone. [90]

Patents are evidence of knowledge claims/IP activity, not manufacturing readiness. EPO and WIPO provide enormous patent corpora, but patents should influence an IP/knowledge layer rather than directly mark factory capability. [91]

Installed capacity is not available qualified capacity. Project state, utilization, yield, maintenance, product mix and qualification all matter. Defense production-readiness practice explicitly treats producibility, tools, test equipment, process maturity and production planning as distinct concerns. [92]

A country's capability is not a scalar. Licensed production, own process, own design, domestic tooling, domestic precursors and competitive production must remain distinct. IndiaBUILD already reflects much of this distinction. [3]

A product does not have a canonical BOM. The published differences between G1, H1 and Figure 03 are already large enough to alter annual actuator, battery and structure requirements materially. [93]

BOM is insufficient without BOP. Two countries can possess identical materials and still differ fundamentally because one lacks the process, machinery, metrology, yield control or qualification required to turn them into the component.

Money is not instantaneous capability. Mining timelines alone make this impossible; announced investment must be transformed through engineering, construction, commissioning and ramp. [94]

Prices cannot remain exogenous for extreme objectives. Once a target becomes a material share of a commodity or equipment market, procurement changes the market being modeled.

Domestic self-sufficiency is not intrinsically optimal. It should be an explicit policy objective or constraint. A resilient architecture may rationally depend on diversified allies rather than fully domestic production.

City-level precision is not automatically sophistication. Model resolution should never exceed evidence resolution.

Important dimensions missing from the original concept

Several omissions are significant enough to change feasibility results.

Inventory and working capital. Shock outcomes depend on days of supply, WIP, strategic stocks and financing of inventories.

Maintenance, repair and spares. Production equipment itself fails. A plant dependent on imported spindle service, control boards or furnace spares is not truly resilient.

Metrology. Precision manufacturing requires the ability to measure tolerances, surface finish, torque, composition and reliability. Metrology is itself an industrial capability.

Quality/yield/scrap. Material input does not equal saleable output. This is especially consequential in semiconductors, batteries, castings and high-precision manufacturing.

Standards, certification and regulatory approval. Technical manufacture can precede legal/customer-qualified manufacture by a long period.

Minimum economic scale. A country may technically be able to make something but not at a sustainable cost below a minimum plant size.

By-products and co-products. Mineral and chemical chains cannot always arbitrarily increase one output without changing co-production or feedstock economics.

Waste treatment and environmental constraints. Refining, plating, semiconductor, battery and chemical capacity may be limited by effluent, waste, land or permitting rather than equipment.

Power quality, not just annual electricity. Fabs, precision plants and chemical facilities need particular reliability/quality characteristics; national TWh availability is not sufficient.

Organizational/tacit capability. Stable high-volume production often relies on process knowledge not visible in patents or machines.

Security/export-control eligibility. A technically available chip or machine may not be legally procurable under a target scenario.

Contractual allocation. Nominal world capacity may already be committed through long-term customer agreements.

Supplier financial health. A small critical supplier can fail financially even if its technology remains sound.

Construction supply chain. Large simultaneous industrial programs compete for EPC capacity, transformers, switchgear, construction equipment, specialty contractors and skilled commissioning teams.

Decommissioning/recycling stocks. Circular supply is constrained by the age distribution of products reaching end of life, not merely recycling percentages.

Industrial coordination. A theoretically feasible collection of projects may fail because sequential investment incentives are misaligned—e.g., no magnet plant without motor demand and no motor plant without magnet supply.

The last point suggests that FORGE should eventually model **coordination failures and anchor-demand policies**, not merely capex totals.

Recommended scope, risks and highest-leverage evolution

The universal vision should remain, but the initial implementation should be deliberately narrower.

Do not build V0 as "ask anything about any physical industry." That architecture would almost certainly become an LLM-generated dependency graph sitting on top of macro datasets—the exact impressive-but-shallow outcome the project is trying to avoid.

Instead:

> **Build a domain-agnostic core ontology and solver, but populate it with highly curated domain-specific knowledge packs.**

V0 should prove that the abstraction generalizes across **two or three substantially different industrial objectives**, not hundreds.

A strong validation portfolio would be:

- one complex discrete electromechanical product: **humanoid robot**;
- one process/material-heavy energy product: **solar PV or batteries**;
- one low-volume/high-complexity aerospace product: **satellite or launch vehicle subsystem**.

This prevents the humanoid ontology from accidentally becoming "the universal manufacturing ontology."

Recommended V0 boundary

Dimension	V0	Later
Geography	India + international supplier countries; country-level macro layers; selected state/site nodes	global site-scale coverage where evidence supports it
Product domains	2–3 curated knowledge packs	expanding library
Decomposition	critical-path/criticality-driven	progressively deeper lazy recursion
BOM/BOP	curated, validated reference architectures	automated assisted extraction + expert validation

Dimension	V0	Later
Capacity	strategic components/processes and major materials	broader facility/line-level capacity
Markets	exogenous baseline demand + reserve constraints	partial equilibrium; selected CGE coupling
Shocks	deterministic scenarios + Monte Carlo parameter uncertainty	stochastic scenario trees, endogenous actor response
Substitution	curated validated alternatives	larger design-search space
Logistics	strategic corridors and landed-cost/capacity constraints	detailed route/inventory/ discrete-event modules
LLM role	query parsing, evidence extraction, candidate generation	same principle; never numerical authority
Output	conditional feasibility, bottlenecks, minimum projects, unknowns	richer dynamic policy/portfolio optimization

Mathematical techniques by subsystem

Subsystem	Recommended core	Avoid initially
Product configuration	constraint satisfaction / typed alternatives	unconstrained LLM BOM generation
Production allocation	LP/network flow	agent-based simulation
New facilities/ capability	MILP fixed-charge capacity expansion	full global CGE
Time	time-indexed MILP + project state machines	one static “capacity” number
Project uncertainty	distributions + scenario analysis	fake point estimates
Shock resilience	scenario re-optimization + inventory	flat downstream multipliers
Competing demand	resource budgets / supply curves	assuming all global supply available
Market feedback	partial-equilibrium approximations	full macro model in V0
Long-run macro feedback	MRIO/GTAP/CGE linkage	engineering inference from IO coefficients
Substitution	discrete validated recipe choice	AI-generated material swapping
Facility/logistics	min-cost capacitated flow	global DES

Subsystem	Recommended core	Avoid initially
Deep transient analysis	targeted discrete-event simulation	simulating every factory
Epistemic uncertainty	interval/probabilistic/Bayesian where justified	one global confidence percentage
Policy portfolio	multiobjective MILP	arbitrary composite “sovereignty score”

Major technical risks

The greatest risk is **false precision**. Rich maps, thousands of graph nodes and exact solver outputs can make weak input assumptions look authoritative.

The second is **ontology explosion**. Industrial reality has near-infinite detail. Lazy expansion and domain packs are the only practical answer.

The third is **identity resolution**: organizations rename, facilities change ownership, products share names and trade classifications do not map cleanly to engineering parts.

The fourth is **data rights**. OpenStreetMap's ODbL, IEA/GEM CC BY data, ecoinvent's paid developer license, patent-service terms and commercial shipment licenses cannot simply be blended into one redistributable database without lineage-aware legal architecture. [95]

The fifth is **maintenance burden**. Capacity, ownership, policies, project status and technology can change continuously. Data freshness must be part of the model.

The sixth is **classification translation**. HS, CPC, ISIC, patent classes and engineering ontologies will never align perfectly.

The seventh is **combinatorial growth**. Multiple product architectures × suppliers × sites × time periods × projects × substitutions can make MILP instances enormous. Hierarchical decomposition and strategic pruning will be required.

The eighth is **endogenous behavior**. Suppliers enter markets, raise prices, reserve capacity and invest in response to demand. Static capacity models become less accurate for trillion-dollar transformation scenarios.

The ninth is **tacit capability**. Some of the most decisive industrial knowledge is not present in public data at all.

The tenth is **model misuse**. Users may treat conditional scenarios as forecasts. Every output must show assumptions and epistemic state prominently.

Highest-leverage changes to the current concept

Reframe the product around buildability, not a "world twin." The term *world model* is intellectually attractive but encourages the impossible promise of full-state knowledge.

Make ProductVariant and ProductionRecipe first-class. Never ask what "a humanoid" needs without selecting an architecture.

Add a formal Product–Process–Resource core. BOM alone cannot establish manufacturability. [96]

Make capability a multidimensional vector. Adopt the strongest aspects of IndiaBUILD's maturity thinking and extend them with yield, capacity, qualification, tooling and upstream dependence. [3]

Distinguish announced, approved, financed, constructed, commissioned, qualified and mature capacity. India's current semiconductor and rare-earth programs demonstrate how dangerous these states are to conflate. [97]

Implement lazy machine-to-machine recursion. Recurse deeply only when capital equipment becomes strategically binding.

Add qualification and metrology as explicit capabilities. They are not annotations.

Add inventories, maintenance and spares. They are essential for shock analysis.

Replace AI substitution with validated alternative recipes. LLMs may suggest candidates; engineering evidence must promote them into the solver.

Build a classification-crosswalk confidence layer. Never silently translate an HS code into an engineering component.

Give UNKNOWN first-class semantics. Missing capacity should not default to zero or to an inferred value.

Separate REPORTED from MEASURED evidence. Company/project announcements should never silently become operational facts.

Build model-result provenance from day one. Retrofitting lineage later will be extremely painful. W3C PROV offers a useful basis. [89]

Make infeasibility explanations a core feature. "Why can't this be done?" may be more strategically valuable than "here is a supply chain."

Add value-of-information optimization. Tell analysts which unresolved evidence is worth paying to collect.

Optimize portfolios of enabling capabilities, not only individual objectives. A precision-machine-tool, metrology or magnet investment may unlock humanoids, aerospace, EVs, industrial robots and defense simultaneously.

Benchmark on multiple industries before claiming domain agnosticism. Domain-agnostic schemas must survive very different production physics.

The most powerful extension discovered in this research

The project can become more valuable than the initial concept by shifting from **single-target planning** to **industrial option-value optimization**.

Instead of only asking:

> "What does India need for one million humanoids?"

FORGE could ask:

> "Which ten capability investments create the greatest increase in India's feasible future production space across robotics, aerospace, energy, semiconductors and advanced machinery?"

IndiaBUILD's capability-adjacency concept provides a natural starting point for this idea. [3] FORGE could add capacity, cost, time and dependency propagation and calculate the **marginal downstream option value** of each enabling capability.

For example, a high-precision machine-tool capability might simultaneously reduce dependence in:

robot actuators
→ aerospace components
→ turbine machinery
→ semiconductor equipment parts
→ medical devices
→ defense systems.

A permanent-magnet capability could unlock multiple motor and generator sectors. A specialty-chemical process could affect batteries, semiconductor manufacturing and advanced materials.

This produces a new class of strategic question:

> **What is the smallest industrial capability portfolio that maximizes the number and value of future objectives a country could execute under plausible shocks?**

That is substantially more differentiated than a supply-chain simulator.

A second high-value extension is the **industrial closure frontier**:

$$\text{cost} \leftrightarrow \text{time} \leftrightarrow \text{domestic value added} \leftrightarrow \text{technology ownership} \leftrightarrow \text{resilience}$$

FORGE could demonstrate, for example, that moving from 70% domestic value addition to 90% is inexpensive, while moving from 90% to full machine-tool/precursor autonomy requires an order-of-magnitude greater investment and several additional years. This is a much more sophisticated view of self-reliance than a percentage localization target.

A third is the **reverse buildability query**:

> "Given today's capabilities, ₹X billion, five years and this import restriction, what is the maximum achievable annual output?"

This dual problem is often more policy-relevant than specifying an arbitrary output target.

A fourth is **portfolio-level competing demand**. Rather than studying humanoids, solar and satellites separately, FORGE could discover that their simultaneous industrial plans compete for the same CNC machine hours, power electronics, specialty alloys or engineering workforce—and optimize shared enabling investments.

A fifth is **observability optimization**:

> "What does the government/company need to measure next in order to make this industrial plan knowable?"

That could become a serious research product in its own right.

Requirements for a future autonomous coding-agent master prompt

A future coding-agent prompt should not begin with UI screens, maps or autonomous web agents. It should begin with **model invariants**.

The master prompt should define FORGE WORLD as:

> **A versioned, evidence-first, unit-safe industrial transformation computation system whose numerical core operates only on typed, validated data and whose language models can assist interpretation and evidence extraction but cannot silently invent production facts, capacities, recipes or feasible substitutions.**

It should require the coding agent to preserve the following non-negotiable principles.

Evidence is separate from model state. Raw/reported claims are immutable objects; normalized model parameters are derived from them.

Every numeric fact has a unit. Dimensional consistency must be mechanically checked. Tonnes of ore, tonnes of contained metal, tonnes of oxide and tonnes of magnet cannot be silently exchanged.

Every claim is time-indexed. Current capability and future/project capability must never share one field.

Every entity is versionable. Plants, projects, organizations, product variants and recipes change over time.

UNKNOWN is a valid database value. The system must not impute unless an explicit estimation method is invoked.

Epistemic class is mandatory. MEASURED, REPORTED, DERIVED, ESTIMATED, MODELLED, ASSUMPTION, SCENARIO, UNKNOWN or CONFLICTED.

LLM outputs are never automatically OBSERVED facts. Extracted claims must retain source evidence and review state.

Recipes are validated computational objects. A production recipe specifies inputs, outputs, yields, processes, equipment, resource requirements and applicability constraints.

Substitutions require an engineering equivalence contract. Performance and transition constraints must be encoded before the solver may select them.

Capability is vector-valued. The schema must not reduce capability to a Boolean.

Capacity is effective and temporal. Nameplate, utilization, yield, uptime, qualification and ramp must remain separately inspectable.

Projects use state machines. Announced/approved capacity does not count as current capacity.

Machine recursion is lazy. Capital-goods decomposition occurs when strategic rules activate it rather than being universally expanded.

The numerical core is deterministic/reproducible given a scenario. Each run stores data snapshot, solver version, scenario, assumptions and random seeds where stochastic methods are involved.

Model runs produce proof graphs. Every important output should expose upstream evidence and derivation.

Optimization must expose binding constraints. Feasibility without explanation is insufficient.

Infeasible and not-adjudicable are first-class outcomes.

Licensing metadata follows the data. A field originating from ecoinvent, OSM, IEA, commercial customs data or a company filing must preserve its source and legal-use metadata; restricted data must not leak into exports. 98

Classification mappings are many-to-many and confidence-bearing. HS↔component or ISIC↔process mappings cannot be silently asserted.

Model resolution cannot exceed evidence resolution. Country-level estimates should not magically become factory-level facts.

Domain-specific knowledge lives in modular knowledge packs. The universal core should not contain humanoid-specific logic.

Every knowledge pack has benchmark tests. For example, multiplying a reference humanoid recipe to one million units must conserve mass, satisfy component counts and expose unmet capacity rather than silently creating supply.

Backtesting is mandatory. Before trusting future transformation plans, the project should replay historical industrial expansions or shocks and ask whether the model would have anticipated known bottlenecks and ramp behavior.

LLMs have bounded roles: natural-language objective parsing, literature search assistance, structured extraction, entity-resolution suggestions, candidate substitution generation and explanation. Numerical quantities, capability status, source promotion and substitution feasibility require deterministic rules and evidence.

The future coding-agent brief should explicitly prohibit several tempting shortcuts:

> Do not generate missing capacities from an LLM.
> Do not treat exports as production capability.
> Do not treat patent ownership as manufacturing capability.
> Do not treat announced projects as available capacity.
> Do not create a canonical BOM where multiple architectures exist.
> Do not use national input-output coefficients as component recipes.
> Do not propagate price shocks with naive downstream multiplication.
> Do not treat “domestic” as equivalent to domestic IP or technological autonomy.
> Do not model global city-level detail from country-level proxies.
> Do not hide assumptions inside generated prose.

The deepest strategic recommendation is therefore to resist building **FORGE WORLD the visualization** before building **FORGE WORLD the epistemic and mathematical contract**.

Public industrial data are strong enough to establish trade, aggregate production, many facilities, mineral and energy systems, macro interdependence, technology signals and large announced projects. [99] They are substantially weaker at the exact layer that determines whether advanced manufacturing works: process-specific effective capacity, yield, qualified suppliers, machine-tool dependencies, proprietary recipes and tacit know-how.

That weakness should not be hidden. It should become part of the product.

The best possible FORGE WORLD is therefore not an omniscient digital twin of Earth. It is an **auditable computational argument about industrial possibility**—one that knows the difference between a tonne of copper and a qualified servo actuator, between an approved fab and a yielding production line, between

assembling a technology and owning it, between buying a machine and being able to reproduce it, between a plausible substitution and an engineering fantasy, and between a missing fact and a fact that an AI merely invented.

If built around those distinctions, its long-run ambition can legitimately exceed conventional supply-chain software: not merely showing **what the industrial world is**, but determining, conditionally and transparently, **what industrial worlds can be built from it, by when, at what cost, with which dependencies, and with what evidence that the plan is real**.

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